

[AI89900] Agentic AI – Research Track

Final Presentation

SeRPO: Segment-level Rubric-based Policy Optimization for LLM Agents

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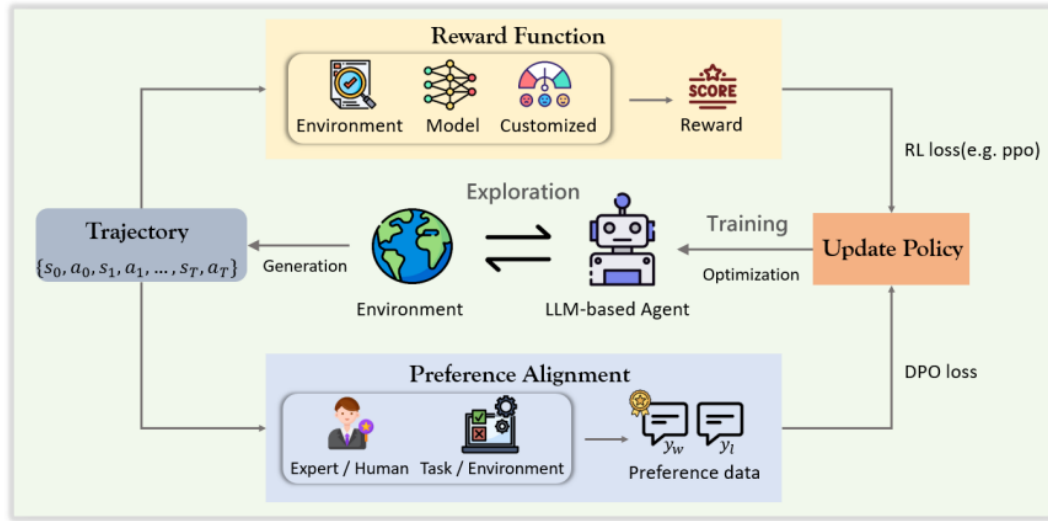
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Conclusion

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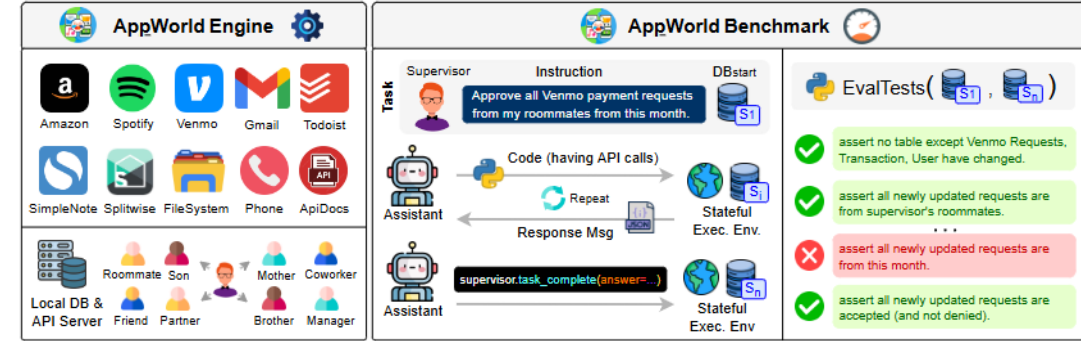
Reinforcement Learning in Agentic AI Reasoning Process

RL in Reasoning

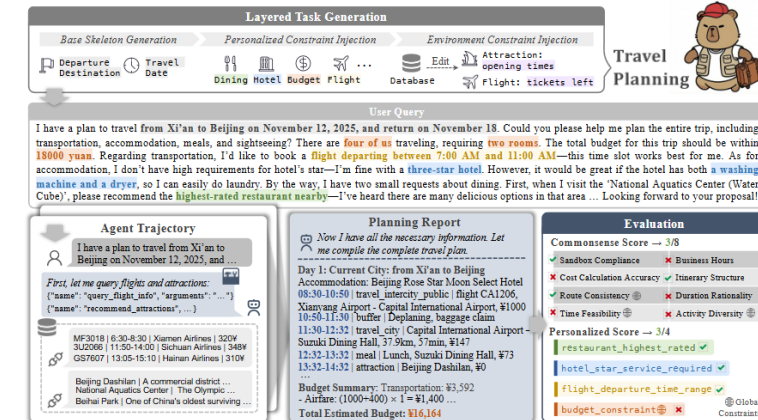


- **Reinforcement Learning** explores optimal strategies by having the agent take direct actions and receive rewards.
- As the performance of agentic AI improves, benchmarks are becoming increasingly long and complex, such as **AppWorld** and **DeepPlanning**

Recent Agentic AI Benchmark



DeepPlanning(2026)



Triggered by the success of DeepSeek-R1, research trend focused on utilizing RL to enhance the LLM's reasoning process, solving complex benchmarks


AppWorld
WHAT THE AGENT IS DOING

Task

Top Artists - Spotify


Request: Give me the 6 artists with the most songs across my Spotify playlists — counting a song once even if it's in several playlists. — same task, two models (Qwen3.5-9B)

9:41


My playlists
6 total


Morning Coffee
playlist

18


Late Night Drive
playlist

24


Gym Mode
playlist

15


Focus Flow
playlist

21


Throwbacks
playlist

19


Rainy Day
playlist

12

3/7 · now: Spotify

WHAT'S HAPPENING NOW

Browse the playlists


Get the user's password
The agent fetches the user's Spotify login from the secure account vault.




Log into Spotify
Signs in to the user's Spotify account with the retrieved password.



3

Browse the playlists
Finds all 6 of the user's playlists. Now it needs the songs inside each one.



4

Collect every song
Opens each playlist and gathers its songs — passing the access token so private lists open too. Duplicates are counted only once.



5

Tally by artist
Groups the songs by artist and ranks them by how many songs each has.



6

Pick the top 6
Selects the 6 artists with the most songs — exactly what the user asked for.




ACTUAL CODE AT THIS STEP

```

all_playlists = []
page = 0
while True:
    res = apis.spotify.show_playlist_library(
        access_token=access_token, page_index=page)
    if not res: break

```

Credit Assignment is the Bottleneck in Long-horizon Agentic RL

RL drives today's LLM agents on long-horizon tasks, but learning depends entirely on **credit assignment**, attributing reward to the actions that earned it. Existing reward designs fail at this in two complementary ways.

① SPARSE REWARD

Sparse rewards → no learning signal

GRPO · DeepSeek-R1 · GiGPO

Reward arrives only at the trajectory's end. In long horizons, successful rollouts are exceedingly rare, so most trajectories return no positive signal and good actions inside failed rollouts get no credit.

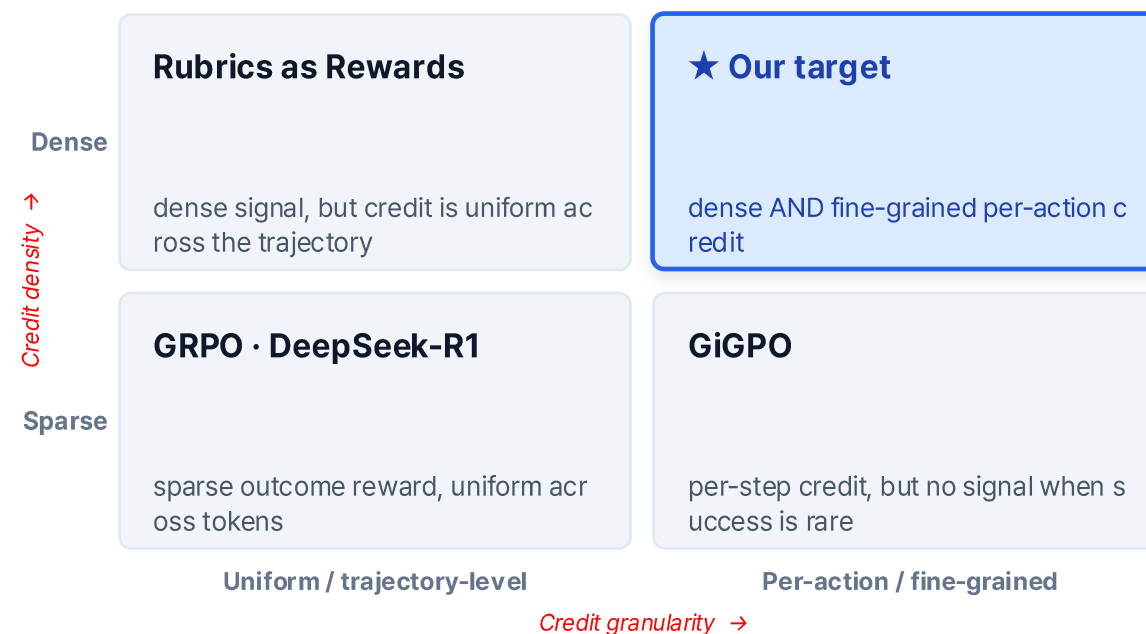
② UNIFORM CREDIT

Uniform credit → no per-action distinction

Rubrics as Rewards

A dense rubric scores the whole rollout at once, so well- and poorly-executed actions receive identical credit, injecting noise straight into the policy gradient.

Where prior methods sit in the credit-assignment space



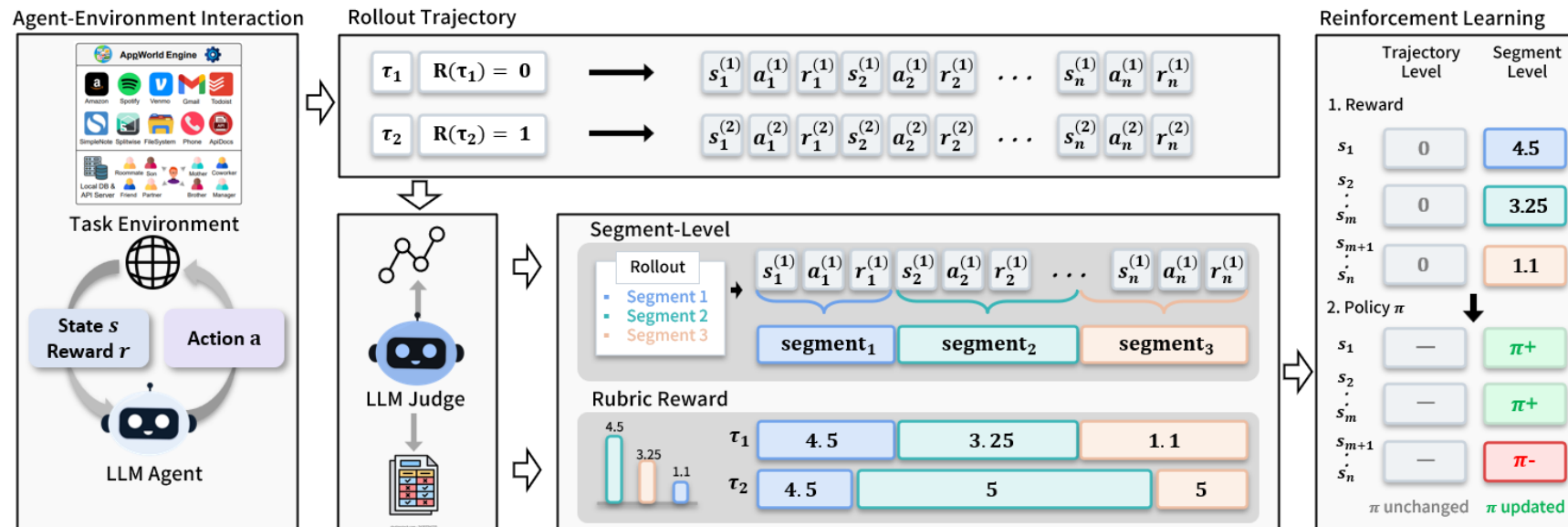
THE GAP

Long-horizon agentic RL needs credit that is **dense enough to learn under rare success** and **fine-grained enough to separate good actions from bad**

Segment-level Rubric-based Policy Optimization for LLM Agents

SeRPO: Segment-level Rubric-based Policy Optimization for LLM Agents

- **Segment-level rubric scoring and policy update**
 - Steps are clustered into **subgoal-aligned segments**, and **rubric-based rewards** are assigned at this intermediate granularity
 - Provides dense, structured credit assignment that is stable to optimize and aligned with the hierarchical nature of long-horizon agent tasks
- **Multi-dimensional rubric-based reward even for failed trajectories**
 - **Assigns non-zero partial credit to failed trajectories** along multiple qualitative dimensions, beyond mere action-format penalties
 - Enables learning even when no successful rollout exists in the batch
 - Rubric evaluation is applied at a finer granularity, distinguishing which parts of a trajectory contributed to or detracted from task progress

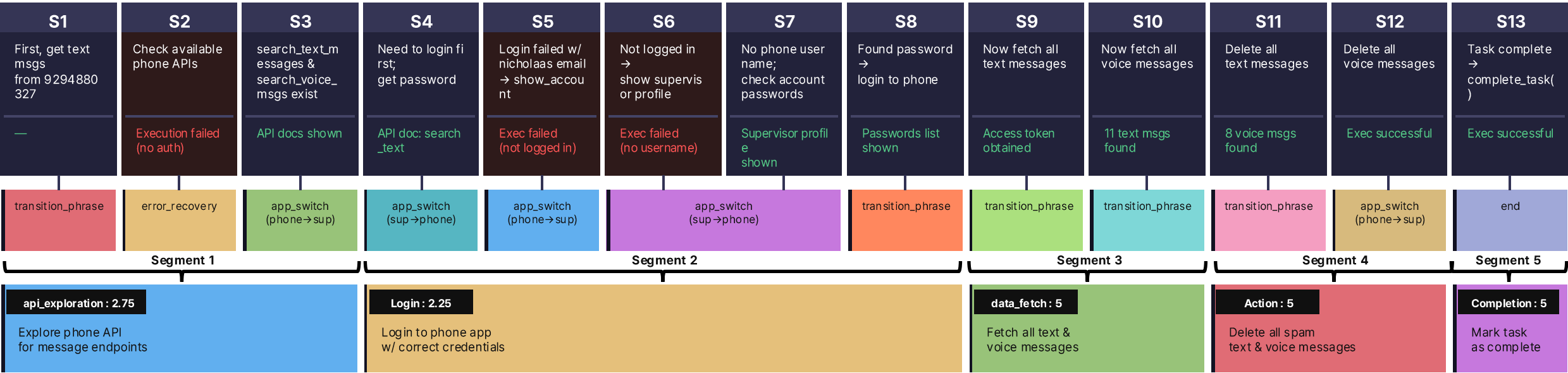


Segment-level Rubric-based Policy Optimization for LLM Agents

Sample: AppWorld train set 90 trajectories

771d8fc_2 | 13 steps | Rule: 12 segs → LLM: 5 segs

Task: "All phone text messages and voice messages from 9294880327 are spam, delete them."



Rubric Reward (1-5)

Dimension	Measurement Signal
Correctness	Count the error responses in the segment's outputs and check whether its final step succeeded.
Efficiency	Count wasted steps = (executed steps) – (steps the subgoal minimally needs), including redundant or repeated identical calls.
Progress	Trace whether the state the segment produced is consumed by a later segment and advances a goal named in the evaluation.
Exploration Fit	Check whether an lookup occurred before the segment's first state-changing action on a previously unseen app.

Experimental Setup



Benchmark: AppWorld

- 9 applications operated through 457 APIs via **free-form code generation**
- code-based action space; even top agents rarely succeed
- Chosen over WebShop / ALFWorld, whose small templated action sets over short horizons let strong agents already saturate



Baselines

- (1) Frozen based LLM (no training)
- (2) GRPO + binary trajectory reward
- (3) GRPO + fractional (test pass ratio) trajectory reward
- (4) GiGPO: binary outcome redistributed to step-level via anchor-state grouping
- (5) SeRPO: LLM judge assigns reward at the segment level



Training Setup

- Base policy: Qwen3.5-9B (ReAct prompting), fine-tuned with LoRA
- GRPO: $G = 8$ rollouts per task
- Judge: GPT-5-mini call at temperature 0



Evaluation

- AppWorld test-normal split
- Metrics: Task Goal Completion (TGC) and Scenario Goal Completion (SGC), both scored by state-based unit tests

Experiment Result

Main Result

- 1

Density is not the bottleneck

- Binary → fractional lifts TGC only +0.7 and leaves SGC unchanged (8.9)
 - A denser signal at the same granularity still cannot say which phase deserves credit
- 2

Step-level helps subtasks, not the task

- GiGPO improves TGC (20.8 vs 19.1) but SGC stays at 8.9, identical to the trajectory baselines
 - Step-level credit is too fine-grained and noisy to represent subgoal structure
- 3

The segment is the sweet spot

- SeRPO is best on both (22.0 / 10.7), the only granularity that lifts long-horizon scenario completion

Method	Reward Type	Credit Granularity	TGC (%)				SGC (%)			
			Avg.	Easy	Med.	Hard	Avg	Easy	Med.	Hard
Base	–	–	17.9	31.6	12.5	9.5	7.1	15.8	6.2	0.0
GRPO	binary outcome	trajectory-level	18.4	38.6	6.2	9.5	8.9	26.3	0.0	0.0
	fractional outcome	trajectory-level	19.1	31.6	10.4	14.3	8.9	15.8	0.0	9.5
GiGPO	binary outcome	step-level	20.8	43.9	8.3	9.5	8.9	21.1	6.2	0.0
SeRPO	LLM-judge	segment-level	22.0	35.1	12.5	17.5	10.7	21.1	0.0	9.5

Ablation Study

Ablation Study: Isolating the Effect of Segment-Level Assignment

- 1

Same reward, helps only at the segment level

- SeRPO reaches 22.0 / 10.7 (TGC/SGC); SeRPO-traj reaches only 17.9 / 5.4
 - Identical scores so the gain is attributable to segment-level assignment, not the rubric
- 2

Averaging gives no gain over Base

- SeRPO-traj does not exceed the untrained base
 - Failed trajectories still contain correct early segments, so averaging shrinks the failure ↔ success gap
- 3

How to assign depends on granularity

- A per-segment signal must stay at the segment level
 - Collapsing it to a single trajectory mean destroys its effect

Method	Reward	Credit Granularity	TGC (%)				SGC (%)			
			Avg.	Easy	Med.	Hard	Avg.	Easy	Med.	Hard
Base	–	–	17.9	31.6	12.5	9.5	7.1	15.8	6.2	0.0
SeRPO-traj	LLM-judge	trajectory	17.9	33.3	8.3	11.1	5.4	10.5	0.0	4.8
SeRPO	LLM-judge	segment	22.0	35.1	12.5	17.5	10.7	21.1	0.0	9.5

Why Segment-Level Assignment Helps

- 1

Survives when the binary outcome is silent

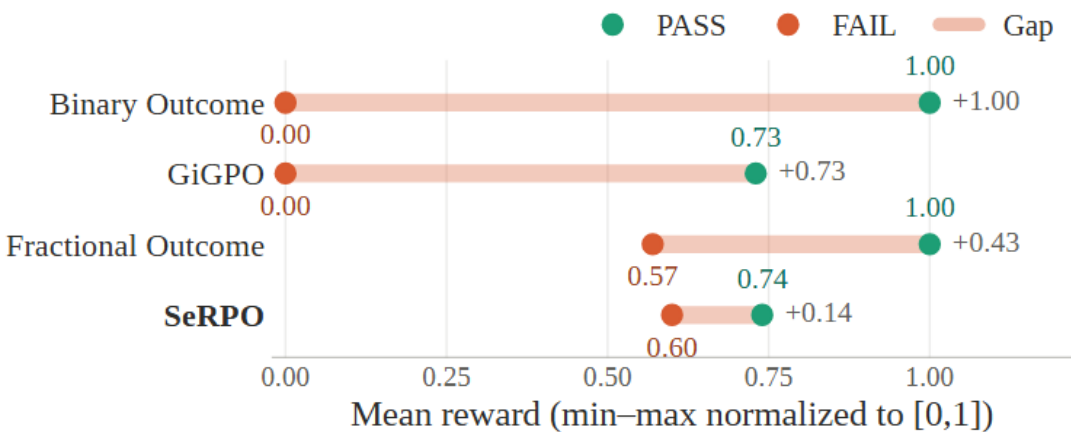
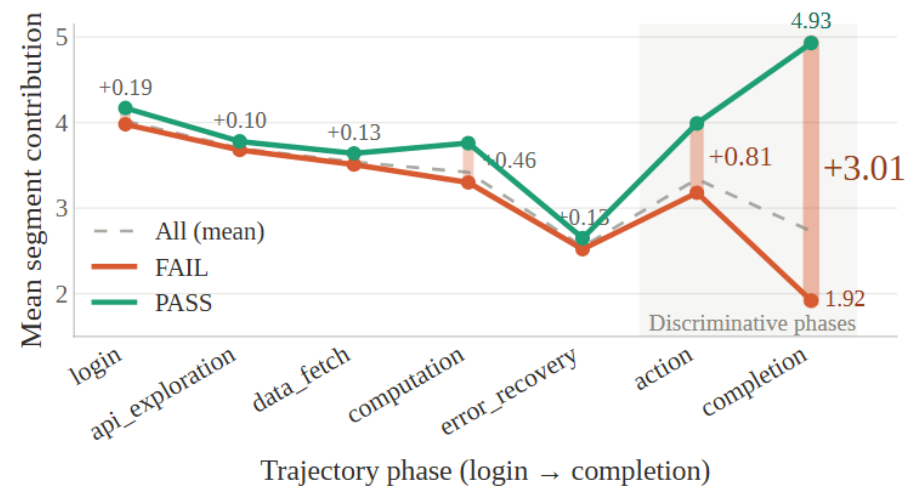
- Across fail groups, segment scores keep non-zero variance, with a pooled within-group std = 1.37 on the 1–5 scale
 - Where binary gives no gradient, the segment still does
- 2

The discriminative signal is concentrated

- Successful and failed rollouts are explored similarly at the initial and the failure surfaces later
 - Completion separates PASS from FAIL almost perfectly
 - The early phases score nearly the same whether the trajectory ultimately passes or fails
- 3

It credits good process, even inside failures

- A failed rollout still earns high marks on clean exploration (api_exploration ≈ 3.68) while the botched completion (1.92) is penalized
 - SeRPO's PASS–FAIL trajectory gap is only 0.14



Analysis

Training on Fail-Only Trajectories

- 1

Outcome rewards collapse to no signal
- 2

Fractional survives only partially

A task-pass-ratio reward works only where rollouts differ in partial completion, and dies on the subset where all rollouts pass zero tests
- 3

SeRPO learns throughout

Outcome-independent segment scores retain non-zero variance even on all-fail groups, so SeRPO alone improves over Base (20.2 / 10.7) using failed data only

Method (fail-only)	TGC (%)				SGC (%)			
	Avg.	Easy	Med.	Hard	Avg.	Easy	Med.	Hard
Base	17.9	31.6	12.5	9.5	7.1	15.8	6.2	0.0
GRPO (binary)	No Learning Signal							
GiGPO	No Learning Signal							
GRPO (fractional)	18.4	35.1	8.3	11.1	5.4	15.8	0.0	0.0
SeRPO	20.2	36.8	12.5	11.1	10.7	21.1	6.2	4.8

Case Study : Step-Level vs Segment-Level

Segment-Level Advantage over Step-Level

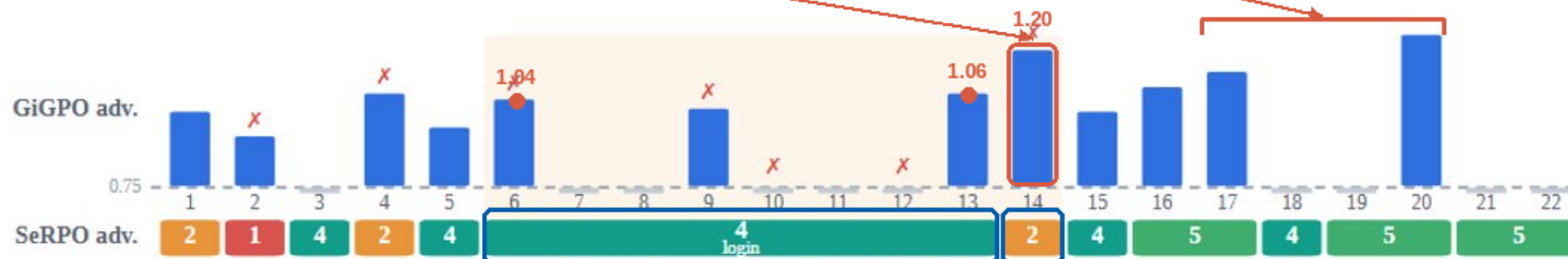
GiGPO — two failure modes of step-level credit

1 Unfaithful to task progress

GiGPO gives the failed 401 retry (step 14) the window's highest score (1.20), and ranks a failed login (step 6, 1.04) on par with the successful one (step 13, 1.06) — its bonus tracks which observation recurs, not task success.

2 Unstable at the step unit

Within one alarm-update subgoal, jointly-needed steps score very differently — some spike above 1.0 while their neighbours sit at the 0.75 baseline.



SeRPO — three properties of segment-level credit

✓ Faithful

Steps 6–13 are wrapped into one 'authenticate' subgoal ($c=4$); the wasteful retry (step 14) is isolated as its own low-credit segment ($c=2$).

✓ Interpretable

Each segment's score maps to a subgoal, so a low-scoring segment points to exactly where credit was mis-assigned or could be hacked.

✓ Stable

Each subgoal gets one consistent value, removing the per-step variance that makes GiGPO's credit scatter across steps of identical role.

Conclusion

Summary

We propose **SeRPO**, which clusters steps into **subgoal-aligned segments** and scores them with a multi-dimensional LLM rubric, filling the gap between step-level credit and trajectory-level rewards

Key Takeaways

1 Segment is the sweet spot

The only granularity that lifts long-horizon scenario completion SGC 10.7 vs 8.9 for every trajectory/step baseline, and best TGC at 22.0

2 Granularity, not density

A denser trajectory reward barely moves the score, and collapsing segment rewards to one trajectory mean erases the entire gain

3 Learns even from failures

Outcome-independent segment scores keep signal on all-fail batches, so SeRPO improves over base using failed data only

Limitations & Future Work

- Evaluated only on AppWorld. generalization to diverse LLM-agent benchmarks (DeepPlanning) is left to future work
- We leave the evaluation of the rubric generator and the alternating-RL stage to future work

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